

CARDIOVIZ XR

48-Hour Medical Animation Challenge - 1-Page Creative Breakdown

1. Creative Concept

Concept: "From Silent Damage to Sudden Blockage." A cinematic vessel cutaway makes an invisible cardiovascular process understandable to a non-medical viewer. Distinct visual cues identify endothelial injury, ApoB particles, plaque, narrowing and thrombus formation.

2. Narrative Flow

Stage	Visual Story	Comprehension Goal
01 - Endothelial injury	Healthy arterial lining develops localized micro-damage.	Show the first invisible change.
02 - ApoB retention	ApoB-containing particles	Stimulus where early plaque begins.
	Plaque enlarges and the artery progressively narrows.	Make progression & space loss obvious.
03 - Plaque growth	Unstable plaque ruptures and a thrombus rapidly forms.	Explain the final rapid events.
- Rupture + clot		

3. Lighting & Texturing

Lighting: Soft clinical key light with controlled rim lighting reveals vessel geometry; localized highlights emphasize plaque growth and rupture.

Materials: Semi-translucent arterial tissue with subtle specular response, with clearly differentiated particle, plaque and clot materials. The look prioritizes medical clarity over unnecessary photorealism.

4. Animation & Pacing

The camera progresses from a wider vessel view into closer cross-sectional detail as disease advances. Transitions remain continuous: injury -> retention -> plaque growth -> rupture -> clot. Short stage labels are used only where they improve comprehension.

5. Asset & Production Notes

Toolchain: Three.js WebGL Engine, Custom GLSL Shaders, React 19, TypeScript, Vite, Motion.

Original work: Vessel geometry, plaque meshes, particle physics, clot system, camera and lighting.

Third-party assets: None. Custom procedural shaders and instanced geometry.

6. Generative-AI Disclosure

Generative AI (Google AI Studio & Gemini models) was utilized as an assistive tool for code generation, UI scaffolding, educational copy synthesis, and storyboard planning. Core 3D shaders and particle systems execute via client-side WebGL.

7. Patient-Comprehension Goal

After one viewing, a non-medical viewer should understand the progression from healthy vessel -> endothelial damage -> ApoB/plaque accumulation -> narrowing -> plaque rupture -> thrombus formation.